

# Yaseen Raihana S

Coimbatore, Tamil Nadu, India | +91 77085XXXXX | yaseenraihana87@gmail.com  
[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [HackerRank](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

Computer Science Engineering (AI & ML) student with a strong foundation in Java, Python, Spring Boot, React, SQL, and Data Structures. Skilled in building full-stack applications and applying Machine Learning, Computer Vision, and Deep Learning techniques to solve real-world problems. Experienced in developing end-to-end ML pipelines, REST APIs, and model evaluation, with a strong problem-solving mindset and a passion for AI/ML-integrated software systems.

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, C++, JavaScript

**Frameworks & Libraries:** Spring Boot, React, TensorFlow, Keras, OpenCV, Scikit-learn, Pandas, NumPy

**Web Technologies:** HTML, CSS, REST APIs, Streamlit

**Databases:** MySQL, SQL, DBMS

**Developer Tools & Cloud:** Git, GitHub, AWS (Basic)

**Core Concepts:** Machine Learning, Deep Learning, Computer Vision, EDA, Model Evaluation, Data Structures, OOP, Problem Solving

## EDUCATION

**Sri Krishna College of Technology, Coimbatore**

B.E., Computer Science Engineering (Artificial Intelligence & Machine Learning)

**2023 – 2027**

CGPA: 7.98/10

## INTERNSHIP

**AI/ML Intern — Skill Forge**

**Duration: 2 Months**

- Completed a hands-on internship in Python, Artificial Intelligence, and Machine Learning at Skill Forge.
- Worked with Python libraries such as Pandas and NumPy to clean, preprocess, and analyze datasets for machine learning tasks.
- Built and evaluated machine learning models under mentor guidance, gaining exposure to the end-to-end ML workflow.

## PROJECTS

**FruitScan – AI-Based Fruit Freshness Detection**

Python, OpenCV, Scikit-learn, Streamlit, Random Forest

- Developed a computer vision application to classify fruits as Fresh or Rotten using Python, OpenCV, and Scikit-learn.
- Built an end-to-end machine learning pipeline including image preprocessing, feature extraction, model training, evaluation, and real-time prediction through a Streamlit interface.
- Engineered ~103-dimensional feature vectors using HSV color histograms, Laplacian texture variance, and BGR channel statistics to improve classification performance.
- Trained a Random Forest classifier (100 decision trees) with an 80:20 train-test split and evaluated performance using accuracy, precision, recall, F1-score, and confusion matrix.
- Designed the application to support both synthetic and real-world Kaggle datasets, enabling future extension to deep learning models such as MobileNetV2.

**Diabetic Retinopathy Detection using Deep Learning**

Python, TensorFlow, Keras, InceptionV3, OpenCV, Scikit-learn, Streamlit

- Designed and trained a multi-class classification model using transfer learning on InceptionV3 to detect five stages of diabetic retinopathy (No\_DR, Mild, Moderate, Severe, Proliferative\_DR) from retinal fundus images.
- Implemented an end-to-end image preprocessing pipeline using OpenCV and PIL, including normalization and augmentation, to prepare fundus images for model training.
- Trained and fine-tuned the InceptionV3 architecture using TensorFlow/Keras for multi-class classification of retinal fundus images.
- Built an interactive Streamlit web interface enabling users to upload fundus images and receive real-time predictions with confidence percentages for each class.

**Event Management Web Application**

React, Spring Boot, SQL

- Built a full-stack event management system with complete CRUD functionality using React and Spring Boot.
- Designed a dynamic, component-based front-end in React for event creation, browsing, and management.
- Developed a RESTful backend with Spring Boot, integrated with an SQL database for persistent, structured data storage.

## CERTIFICATIONS

- AWS Cloud Practitioner Essentials — Amazon Web Services
- AWS Certified Developer – Associate — Infosys
- Microsoft Azure Fundamentals — Microsoft
- Data Analytics — Maiyyam
- Cloud Computing — NPTEL